Deleted irreps as unknowns. Axial torsion, tensor torsion, and traceless nonmetricity are not coordinates of H_{full}. Treating them as solved variables changes the domain rather than enlarging the kernel computation.

No additional algebraic factor of $\det H_\perp$ appears. The regular proof is not extended through S2–S4.

J Minimal deformation control of the residue fingerprint

The following deformations test the residue fingerprint; the horizontal system defined by (3.7) is left unchanged. A local deformation is called *symmetry-compatible* when it is a diffeomorphism scalar constructed from the projectively invariant horizontal one-forms, at the same algebraic and derivative order as the undeformed quadratic density, and independent of the vertical coordinate V.

Write $x = (q, u, t)$ and isolate the Palatini block P, mixing B, and residue coefficient C in the $\frac{1}{2}x^{\mathsf{T}}Hx$ normalization,

$$B = f_1 \begin{pmatrix} -2/3 \\ 2/3 \end{pmatrix}, \qquad C = f_1\big(-\tfrac{4}{3}\big) + 2f_2, \tag{J.1}$$

so that the t-Schur coefficient is $S_t = C - B^T P^{-1} B$. The undeformed value is $S_t(0,0) = 2f_2$, equivalently $\Delta_2 = 0$.

Consider the nearby symmetry-compatible quadratic deformation

$$\delta L_{\text{quad}} = \alpha\, q \cdot t + \beta\, u \cdot t, \tag{J.2}$$

with independent symbolic coefficients α, β. This shifts $B \mapsto B + \delta B$ with $\delta B = (\alpha, \beta)^{\mathsf{T}}$. Then

$$S_t(\alpha, \beta) = C - (B + \delta B)^T P^{-1} (B + \delta B). \tag{J.3}$$

Using (5.17) one finds $B^T P^{-1} \delta B = \frac{8}{3}\alpha + \frac{2}{3}\beta$ and

$$\delta B^T P^{-1} \delta B = f_1^{-1}\Big(-\tfrac{4}{3}\alpha^2 + \tfrac{16}{3}\alpha\beta + \tfrac{11}{3}\beta^2\Big). \tag{J.4}$$

Hence

$$\delta S_t := S_t(\alpha, \beta) - S_t(0,0) = -\tfrac{16}{3}\alpha - \tfrac{4}{3}\beta - f_1^{-1}\Big(-\tfrac{4}{3}\alpha^2 + \tfrac{16}{3}\alpha\beta + \tfrac{11}{3}\beta^2\Big). \tag{J.5}$$

For each $f_1 \neq 0$, the zero locus of (J.5) is a plane conic of measure zero in the (α, β) plane. Off that conic, $\delta S_t \neq 0$, so Δ_2 is corrected. In particular, the line $\beta = -4\alpha$ kills the linear part and yields $\delta S_t = -36\alpha^2/f_1$, which vanishes only at $\alpha = 0$.

The undeformed t-source Schur correction is $J_{t,\text{eff}} = -\partial f_3$, equivalently $\Delta_3 = 0$. With (J.2) one obtains the extra term

$$J_{t,\text{eff}}(\alpha, \beta) = -\partial f_3 + \frac{\partial f_1}{f_1}(4\alpha + \beta). \tag{J.6}$$

Thus Δ_3 remains zero if and only if $\beta = -4\alpha$. Combined with (J.5), both $\Delta_2 = \Delta_3 = 0$ hold simultaneously only at $(\alpha, \beta) = (0, 0)$.

A minimal symmetry-compatible linear-source deformation $J_N \mapsto J_N + (\varepsilon_q, \varepsilon_u)$ produces, at $\delta B = 0$,

$$\delta J_{t,\text{eff}} = -\big(\tfrac{8}{3}\varepsilon_q + \tfrac{2}{3}\varepsilon_u\big). \tag{J.7}$$

This vanishes if and only if $\varepsilon_u = -4\varepsilon_q$. A generic source deformation therefore yields a nonzero t-source Schur correction.

The fingerprint $\Delta_2 = \Delta_3 = 0$ is explicitly non-generic within the stated deformation family. These diagnostic deformations do not modify the defining equations of the model.